User Activity Monitoring and Anomaly Detection

Kim Nguyen

Department of Computer Science and Computer Engineering (CECS), California State University, Long Beach

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Professor Samuel Addington

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Abstract

In our digital age of technology, our world is defined by ubiquitous data with over 90% of the world's data having been created in just past two years, growing our ecosystem from gigabytes to a zettabyte-scale. All of this data comes from our smart home devices to industrial machines, constantly collecting and sharing our daily actions like using Global Positioning System (GPS), social media, and online browsing create a constant flow of information from over 5.78 billion unique mobile users (Digital, n.d.). The rapid development of technology and data generation has transformed anomaly detection from a reactive, manual process into a an automated, real-time capability (Digital, n.d.). Traditional rule-based methods are no longer sufficient to handle the volume and variety of modern data, thus making advanced AI techniques essential for detecting anomalies across large, high-dimensional datasets (Digital, n.d.). This project addresses the problem of detecting abnormal user activity within systems by applying automated anomaly detection techniques to activity logs. By analyzing patterns like login times, file access frequency, and device usage, the system aims to identify behavior that significantly deviates from established user baselines to help identify potential security risks. The goal of this project is to design and evaluate an automated system that identifies deviations from normal user behavior in enterprise activity logs using anomaly detection methods.

The primary assets that are considered in this project include sensitive organizational files, user accounts and credentials, internal systems and databases, removable media, and other endpoint devices. These assets are important because data misuse or unauthorized access could potentially lead to undesirable outcomes such as financial loss, exposure of sensitive information, overall the disruption of organizational operations, and privacy breaches of personal information. While external attackers such as hackers are an obvious threat, some other potential attackers include compromised internal accounts that behave like legitimate users, negligent insiders who violate policy unintentionally, former employees who still retain access or knowledge of internal systems, and even malicious employees attempting to steal

data. This makes the detection problem challenging because the activity may originate from authorized accounts rather than obvious external attackers.

The primary attack surfaces include authentication systems such as workstation or server logins, file access and data transfer mechanisms, removable media or external device connections, web-based uploads, and the use of privileged accounts (What is, 2025). These points of interaction are particularly important because they represent common pathways through which users access, transfer, or modify sensitive organizational data (What is, 2025). If misused or compromised, these access points may allow attackers or unauthorized users to exfiltrate sensitive information, introduce malicious software, or abuse elevated privileges to gain further access within the system. Shared databases and weak authentication practices alongside unmonitored devices can further increase the organization's attack surface and increase the likelihood of unauthorized access or data exposure (What is, 2025).

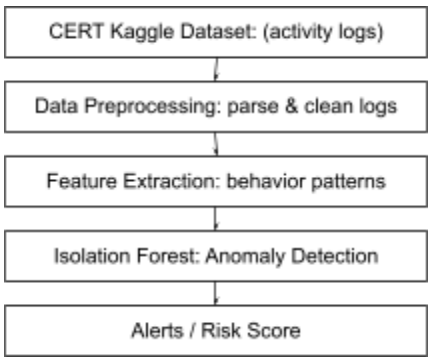
This project assumes that enterprise activity logs are available and contain reasonably accurate records of user actions like login information, file access, device usage, and historical activity data. These things can all be used to establish a baseline of normal user behavior within the organization. Because unusual behavior doesn't always equate to malicious intent, the system's alerts are intended to highlight potentially suspicious activity that may require further investigation rather than automatically labeling actions as attacks.

The planned analytic approach is to develop a behavioral anomaly detection pipeline that models normal patterns of user behavior and identifies deviations from those patterns. The system will analyze activity logs to detect indicators such as logins occurring outside normal work hours, unusual spikes in file access or copying, unfamiliar device usage, a user's access to resources they don't usually interact with, and other sudden deviations from past behavioral patterns. The system will use Isolation Forest, an unsupervised anomaly detection method, to detect abnormal patterns in user activity logs and will be evaluated using several performance metrics (Barnard & Stryker, 2025). Detection rate or recall measures how many suspicious

events are correctly identified. Precision evaluates how many alerts correspond to potentially malicious activity. The false positive rate measures how often normal user behavior is wrongly flagged as suspicious. Alert latency evaluates how quickly suspicious activity is detected after it occurs. Then, user-level anomaly score quality measures the system's effectiveness in ranking higher-risk activities for investigation (Barnard & Stryker, 2025).

The project will use the CERT Insider Threat Detection Dataset on Kaggle (Mrajaxnp, 2023) as the primary data source which contains simulated enterprise activity logs including user logins, file access events, device connections, email activity, and web browsing records. These logs represent both normal user behavior and scenarios involving suspicious activity. The dataset is synthetic and publicly available, meaning it was artificially generated to simulate realistic organizational activity without containing real user information. That makes it safe to use for research purposes. The dataset will be used to extract behavioral features and evaluate the anomaly detection system's ability to identify abnormal user activity.

Analyzing user activity logs may raise privacy and ethical concerns related to monitoring user behavior. To mitigate these risks, this project will use the publicly available CERT Insider Threat dataset from Kaggle, which is synthetic and anonymized and does not contain real user data. In addition, detected anomalies will be treated as indicators for further investigation rather than definitive proof of malicious activity. Below is the architecture that shows how activity logs from the dataset are processed, transformed into behavioral features, analyzed using anomaly detection methods, and then used to generate alerts for potentially suspicious user activity.



Main Body

The system created was developed to automate anomaly detection to analyze user activity logs and identify random deviations from normal behavorial patterns. By analyzing typical behavorial patterns like login activity in combination with device usage and file access, the system will be able to identify potential security risks through the misuse of system resources or unauthorized access. This end-to-end pipeline system consists of 4 main stages which are data ingestion, anomaly detection, alert generation, and result summarization. Developed using Python and Docker, the system can be reproduced and deployed across different environments and can be executed using a single command that enables automated user activity data processing.

The primary data source used in this system was a subset of the CERT Insider Threat dataset from Kaggle for the simulated enterprise activity logs, this project takes the logon, file, and device activity logs. Because of the repository size limitations, a reduced sample of the dataset had to be used for implementation and testing while still being able to maintain variability. The raw log data is initially preprocessed by converting certain fields, like the timestamp fields, into datetime formats that were more structured so that it can be worked with easily. Then, the data is aggregated so that the system could analyze behavorial patterns over time. Using this processed data, the system extracts some features to represent user activity including the login count per day, the amount of logins after-hours, and the number of unique machines that were accessed, the frequency of file access, removable media use, and connection activity for each device. All of these serve as input for the anomaly detection model.

The system applies an unsupervised machine learning algorithm called Isolation Forest to detect anomalous behavior. It isolates data points that differs significantly from most of the other observations, which makes it good for detecting outliers. The model doesn’t need to have labeled data since it is unsupervised, which lets it detect previously unseen or unknown abnormal behavior. The model has a contamination rate of 2% to flag a anomalies in the dataset

to focus on the most unusual activity patterns. In addition to that, the model also gives us an anomaly score for each user's day record that determines whether it was normal or anomalous. Then, those results are used to generate alerts and summaries for us to analyze further.

The system was able to detect anomalies stored in a CSV file and then generate several output items that support the analysis and evaluation that will be used for inspection. For example, the csv file "alerts.csv" contained flagged anomalous events, "metrics.json" summarized total records and anomaly counts, "top_anomalies.csv" listed highest anomaly scores, "anomaly_counts_by_user.csv" summarized anomalies per user, and "anomalies_by_user.png" was an image that showed top anomalous users (see Figure 1). Overall, the results showed us that the model was able to identify a small subset of user activity as anomalous. Users that had higher anomaly counts exhibit deviations like unusual login times or higher activity levels.

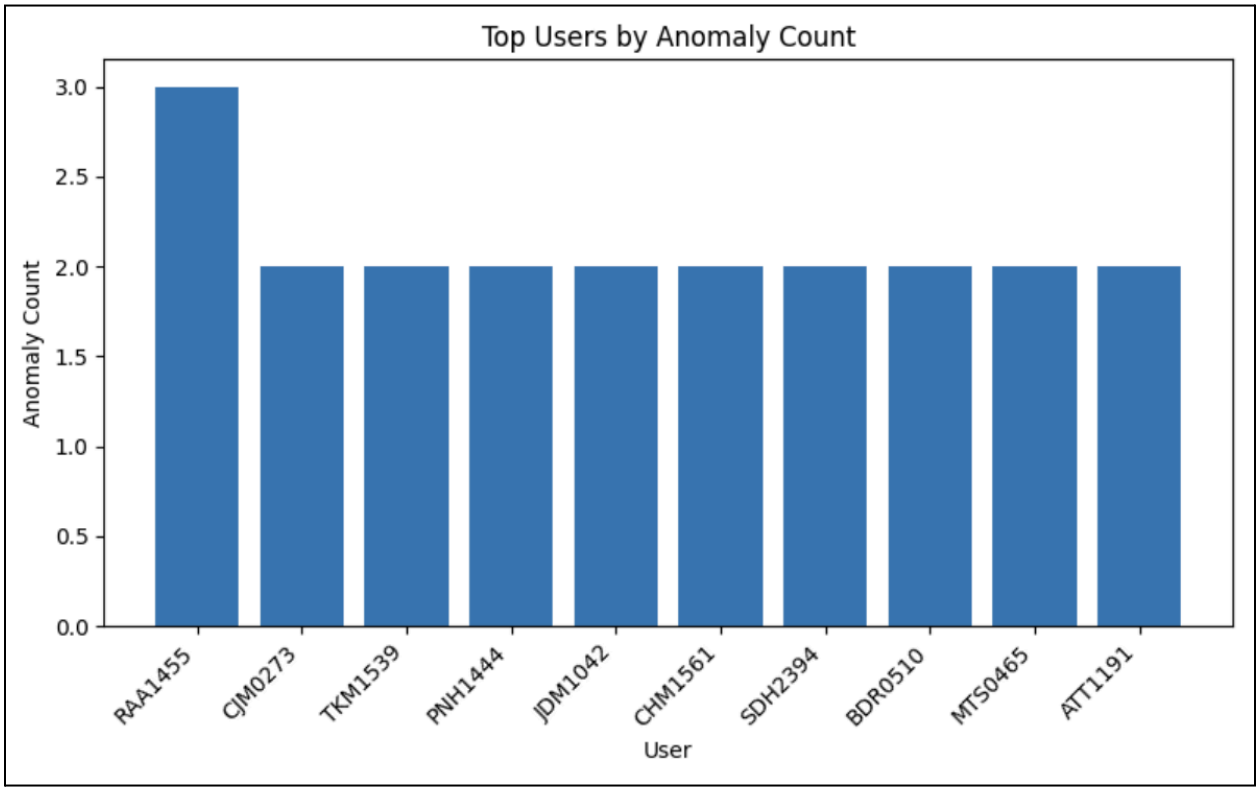


Figure 1

To ensure that the system was reliable, happy-path test was done to verify that the system was able to correctly process valid input data and produce expected outputs (see Figure 2). Another negative and edge-case tests evaluate system behavior under conditions like missing required fields, invalid data formats, and empty inputs. Also, the model implements measures to take into consideration the security and robustness of the system. Input validation makes sure that required data fields are inputted before the processing part begins, which reduces runtime errors or wrong input. The system works only on structured CSV data and doesn't need external network access, ensuring less exposure to external threats.

```
docker compose run --rm -e PYTHONPATH=/app app python -m pytest tests --cov=src --cov-report=term-missing
========================================================================== test session starts ==========================
platform linux -- Python 3.12.13, pytest-9.0.3, pluggy-1.6.0
rootdir: /app
plugins: cov-7.1.0
collected 4 items

tests/test_pipeline.py ....

============================================================================== tests coverage ===========================
_____________________________________________________________ coverage: platform linux, python 3.12.13-final-0 _____

Name                  Stmts   Miss  Cover   Missing
-----------------------------------------------------
src/__init__.py           0      0   100%
src/explore_data.py      13     13     0%   1-18
src/pipeline.py          73     38    48%   22-26, 105-110, 114-165, 169-174, 178
-----------------------------------------------------
TOTAL                    86     51    41%
========================================================================== 4 passed in 3.78s ============================
(.venv) kimnguyen@MacBook-Pro-3 CECS478-Project %
```

Figure 2

The initial evaluation shows that the system could identify a subset of user activities as anomalous. As expected, the detected anomalies tended to have a correspondence with patterns like login times that were out of the ordinary, more file access, or increased device use. This suggests that the features that were selected and the anomaly detection approach are effective in being able to capture deviations from typical behavior. On the downside, the model is an unsupervised one so the data that we received does not actually indicate whether the anomalies actually correspond to malicious actions, so more analysis and model tuning is needed in order to improve accuracy and lessen false positives. Also, as mentioned before, the dataset had to be reduced to repository size constraints.

References

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